

Chapter 105 — Evaluation and Management of Women With Urinary Incontinence and Pelvic Organ Prolapse

IF YOU READ NOTHING ELSE

- **POP (pelvic organ prolapse) diagnosis needs symptoms** — ~50% parous women have anatomical POP.
- **Vaginal bulge** = most POP-specific symptom; most likely to resolve after repair.
- **POP symptomatic at/beyond hymen** (stage ≥ 2).
- **SUI (stress urinary incontinence) index patient pre-surgery:** UA (urinalysis) + PVR (postvoid residual) + objective SUI demonstration (AUA/SUFU [American Urological Association/ Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction] 2023).
- **Cystoscopy, UDS (urodynamics), imaging not routine** in index SUI.
- **POP-Q (POP Quantification) stage II** = leading edge -1 to $+1$ cm; **III** $>+1$ to $<+(tvL [total vaginal length] -2)$; **IV** $\geq+(tvL-2)$.
- **Occult SUI** = one-third of vaginal POP repairs; **SUI coexists** in ~40% POP.
- **Urethral hypermobility** = $>30^\circ$ from horizontal.
- **Translabial US (ultrasound)** = only reliable mesh imaging; **MRI (magnetic resonance imaging)** = standard for urethral diverticulum.
- **Single PVR unreliable** — repeat; no agreed cut-off (30–200 mL).

DEFINITION AND IMPACT

- **PFDs (pelvic floor disorders)** = UI (urinary incontinence), FI (faecal incontinence), POP, LUT (lower urinary tract) sensory/emptying, defecatory dysfunction.
- **POP, UI or FI** in ~1 in 4 women; rising (age, obesity).
- **UI (ICS, International Continence Society):** complaint of involuntary loss of urine.
- **Commonest subtype:** isolated SUI $>$ UUI (urgency UI) $>$ MUI (mixed UI); UUI rising ≥ 60 years.
- **UI all types $\uparrow$:** age, obesity, smoking, comorbidity, postmenopausal HRT (hormone replacement therapy).
- **Vaginal delivery $\rightarrow$ SUI;** caesarean/activity $\rightarrow$ less UUI.
- **Pregnancy UI:** ~10% first $\rightarrow$ ~33% third trimester.
- **Black women:** more UUI/DO (detrusor overactivity); White/Mexican-American: more SUI.
- **Lifetime surgery risk by 80:** SUI 13.6%, POP 12.6%.
- **Commonest POP site:** anterior $>$ posterior $>$ apex.
- **POP recurrence risk:** younger age, POP-Q stage 3–4.
- **POP protective:** caesarean delivery, smoking.
- **Care-seeking:** only 18–50%; delay 1 to ≥ 5 years.
- **Occult SUI (ICS)** = SUI only on POP reduction.
- **Insensible UI:** aware of leak, not how/when.
- **DIAPPERS (Resnick):** transient/reversible UI causes.

DIAGNOSTIC EVALUATION

- **SUI = urethral; UUI = bladder; MUI both.**
- **MUI:** treat most bothersome component first.
- **Positive SUI history $\rightarrow$ urodynamic SUI** in ~3/4.
- **Splinting $\rightarrow$ high-grade POP; high-grade anterior POP $\rightarrow$ obstructed voiding.**
- **POP repair may improve OAB (overactive bladder)** — not always; stage OAB treatment.
- **De novo SUI:** no ICS definition; after POP surgery, no preop occult SUI.
- **Urinary symptoms may precede neurological diagnosis.**
- **Local oestrogen** may improve UI; **systemic** (incl. OCP, oral contraceptive pill) may worsen.
- **Frail elderly (6th ICI, International Consultation on Incontinence):** catheter drainage if PVR 200–500 mL, then reassess.
- **Frail elderly red flags $\rightarrow$ refer:** pain, haematuria, recurrent UTI (urinary tract infection), pelvic mass, irradiation, prolapse beyond introitus, fistula.
- **Do not treat asymptomatic bacteriuria.**

- **α -blockers $\rightarrow$ $\downarrow$ outlet $\rightarrow$ worse SUI.**
- **Cholinesterase inhibitors $\rightarrow$ $\uparrow$ contractility $\rightarrow$ UUI.**
- **ACE (angiotensin-converting enzyme) inhibitors $\rightarrow$ cough $\rightarrow$ SUI.**
- **SGLT2 (sodium-glucose cotransporter 2) inhibitors $\rightarrow$ polyuria, $\uparrow$ UTI; lithium $\rightarrow$ diabetes insipidus.**
- **Glitazones $\rightarrow$ oedema $\rightarrow$ nocturnal polyuria.**

PHYSICAL EXAMINATION

- **Frailty assessment:** preoperative, age ≥ 65 .
- **Fried Frailty Index:** 5 domains incl. grip, walking speed.
- **Clinical Frailty Scale:** 9-point pictorial, rapid.
- **Atrophic:** pale, flat, dry, no rugae; **oestrogenised:** pink, rugated.
- **SUI demonstration:** comfortably full bladder; stand if supine negative.
- **CST (cough stress test) at half functional capacity** (largest diary void) $>$ full bladder for specificity.
- **POP present $\rightarrow$ reduce anterior wall** before SUI testing.
- **Q-tip:** hypermobility $>30^\circ$; now trials/teaching only.
- **VUME** (visual urethral mobility examination) = routine; vaginal swab $\equiv$ Q-tip.
- **POP exam:** lithotomy + standing (foot on step).
- **Posterior blade expelled on strain $\rightarrow$ level I** (cardinal–uterosacral) loss $\rightarrow$ vault prolapse.
- **Blade stays $\rightarrow$ isolated rectocele/enterocele.**
- **DeLancey:** levels I–III of vaginal support.
- **Baden-Walker:** grades 1–4 relative to hymen; ordinal.
- **POP-Q (1996):** 9 points, cm, relative to hymen, on Valsalva.
- **Aa/Bp:** 3 cm inside hymen, range -3 to $+3$.
- **Ba/Bp:** mobile leading points, -3 to $+tvL$.
- **D** = posterior fornix; absent after hysterectomy.
- **tvL** measured with prolapse reduced.
- **Stage 0:** Aa, Ap, Ba, Bp -3 ; C/D $\leq -(tvL-2)$.
- **Stage I:** leading edge < -1 cm.
- **BCR (bulbocavernosus reflex):** S2–S4; present 70% women vs 100% men.
- **Anal tone** = S2–S4.
- **IUGA (International Urogynecological Association)/ICS 2011:** prominence = no epithelial breach; separation = breach.
- **Extrusion** = out through tissue; **perforation** = into viscus.
- **Sinus tract** = fistulous track, no visible mesh.
- **Mesh tenderness** documented even without exposure.
- **Genital exam** triggers flashbacks in ~50% of trauma survivors.

- **Child sexual abuse:** 1 in 4 women, 1 in 13 men.
- **Chaperone:** always honour request; use even with companion present.

SUPPLEMENTAL EVALUATION

- **UA + PVR:** most UI patients.
- **AUA OAB:** UA required; PVR, culture, diary optional.
- **AUA OAB:** no UDS, cystoscopy or US in uncomplicated patient.
- **Extra SUI tests if:** diagnostic doubt, SUI not shown, neurogenic, haematuria/pyuria.
- **Extra SUI tests if:** urgency-predominant MUI, high PVR, voiding dysfunction.
- **Extra SUI tests if:** POP-Q ≥ 3 without SUI on reduction.
- **ICIQ-UI SF (ICI Questionnaire, UI short form)** correlates with 1-h and 24-h pad tests.
- **QUID (Questionnaire for UI Diagnosis):** 6 items, SUI vs UUI. **OAB-V8 (OAB awareness tool):** primary-care screen.
- **PFDI (Pelvic Floor Distress Inventory)** = POP symptoms; **PFIQ (Pelvic Floor Impact Questionnaire)** = POP QoL (quality of life) impact.
- **ICIQ bladder diary:** 3 days.
- **1-h pad test:** 500 mL in <15 min; walk 30 min; restart if voids.
- **1-h severity:** 1–10 g mild; 11–50 moderate; >50 severe.
- **24-h severity:** 4–20 g mild; 21–74 moderate; ≥ 75 severe.
- **Non-urine gain:** <1.4 g (1-h); <4.4 g (24-h).
- **24-h test** more reproducible; does not predict urodynamic SUI.
- **Pad counts** don't correlate with pad weight.
- **Oral phenazopyridine** 100–200 mg tds $\rightarrow$ urine orange.
- **Double-dye:** orange tampon = ureterovaginal; blue = vesicovaginal.
- **PVR:** US preferred over catheter.
- **Asymptomatic women:** almost all PVR <100 mL.
- **Cystoscopy:** haematuria, urgency/irritative LUTS (lower urinary tract symptoms), prior mesh, radiation.
- **Low-risk pre-op cystoscopy** not required.
- **Trabeculation** $\rightarrow$ outlet obstruction and/or DO.

URODYNAMIC STUDIES — IMAGING

- **No imaging** in uncomplicated UI.
- **VCUG (voiding cystourethrogram):** VUR (vesicoureteric reflux), urethral diverticulum, subtle SUI.
- **Upper-tract US if:** retention, severe POP, fistula, neurogenic detrusor.
- **CT (computed tomography)/MRI poor for mesh** unless infected.
- **Urethral diverticulum** — 3 Ds: dysuria, dribbling, dyspareunia (often absent).
- **HMO (H-line, M-line, organ prolapse; MRI):** H = hiatus width; M = levator descent from PCL (pubococcygeal line).
- **MRI POP** = organ beyond H-line.
- **Dynamic MRI:** symptoms $\gg$ findings, prior repair, complex.

CLASSIC TRAPS

- **Trap:** anatomical POP = disease — **Actually:** symptoms required.
- **Trap:** POP repair cures UI — **Actually:** distinct diagnoses; SUI may be unmasked.
- **Trap:** older age $\rightarrow$ POP recurrence — **Actually:** younger age.
- **Trap:** smoking promotes POP — **Actually:** protective for primary POP.
- **Trap:** SUI test with POP unreduced — **Actually:** prolapse kinks urethra, masks SUI.
- **Trap:** absent BCR in woman = neuropathy — **Actually:** absent in 30% normal women.
- **Trap:** pad count = severity — **Actually:** poor correlation with pad weight.
- **Trap:** one low PVR excludes retention — **Actually:** repeat; single value unreliable.
- **Trap:** CT/MRI to find mesh — **Actually:** translabial US.
- **Trap:** α -blockers harmless in women — **Actually:** worsen SUI.